Project Report

Smart Biometric Face Recognition Tracker

Bachelor's Degree (CSE) | KIIT University

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01 / Introduction

The Need for Smart Attendance

Why Attendance Matters

Attendance tracking is fundamental in educational institutions for monitoring student participation, ensuring academic integrity, and maintaining institutional accountability. However, traditional methods create significant operational challenges.

Challenges of Traditional Methods

Manual roll calls are time-consuming, prone to human error, and susceptible to proxy attendance where one student marks attendance for another. These inefficiencies waste valuable class time and compromise data accuracy.

Key Issues

- **Time Inefficiency:** Manual processes consume 5-10 minutes per class
- **Proxy Attendance:** Students can easily manipulate the system
- **Data Errors:** Manual entry leads to inaccuracies in records
- **Lack of Security:** No verification mechanism in place

Our Solution

A **Face Recognition-based Attendance System** that automates the entire process, eliminates manual errors, prevents proxy attendance, and operates securely without requiring physical contact.

02 / Problem Statement

Critical Challenges

Time-Consuming

Manual attendance marking takes 5-10 minutes per class session, significantly reducing effective teaching time. In a typical semester, this accumulates to hours of lost instructional time.

Impact: Reduced learning efficiency and instructor productivity

Proxy Attendance

Traditional methods allow students to mark attendance for absent classmates. This undermines academic integrity and makes it impossible to accurately track actual student participation.

Impact: Compromised attendance data and institutional credibility

Contact-Based Systems

RFID cards and fingerprint scanners require physical contact, raising hygiene concerns especially in post-pandemic scenarios. These systems also have hardware dependencies and maintenance costs.

Impact: Health risks and increased operational expenses

Required Solutions

- **Contactless System**
  No physical interaction required
- **Accurate Identification**
  Prevent proxy and false entries
- **Offline Capability**
  Work without internet dependency

03 / Objectives

Project Goals

01

Face Recognition System

Develop a robust face recognition-based attendance system that automatically identifies students and marks their attendance with high accuracy.

Primary Goal

02

Mask Detection

Implement mask detection capability to identify whether a person is wearing a face mask, ensuring compliance with health protocols.

Health Safety

03

Accurate Marking

Ensure precise attendance marking with confidence level verification to eliminate false positives and maintain data integrity.

Data Integrity

04

Offline Operation

Enable the system to function completely offline without internet dependency, ensuring reliability and accessibility in all environments.

Independence

05

Secure Storage

Store attendance data securely in both CSV and MySQL database formats, providing redundancy and easy data management capabilities.

Data Security

04 / Technologies

Technology Stack

Python

Primary programming language chosen for its extensive libraries, ease of use, and strong community support in computer vision applications.

OpenCV | NumPy | Pandas

Face Detection

Haar Cascade classifier used for rapid face detection in real-time video streams with pre-trained models optimized for frontal face detection.

Real-time | Efficient | Lightweight

Face Recognition

LBPH (Local Binary Patterns Histograms) algorithm for robust face recognition that works well with varying lighting conditions and angles.

LBPH | Robust | Scalable

Database

MySQL relational database for structured storage of attendance records with support for complex queries and data analysis.

MySQL | Structured | Reliable

Libraries

Essential Python libraries for image processing, numerical operations, and data manipulation to support the complete pipeline.

- **OpenCV:** Computer vision and image processing
- **NumPy:** Numerical computations
- **Pandas:** Data manipulation and analysis

Hardware

Minimal hardware requirements making the system accessible and cost-effective for educational institutions.

- **Webcam:** HD camera for video capture
- **Computer:** Standard PC with 4GB+ RAM
- **Storage:** Local disk for dataset and models

05 / Architecture

System Architecture

Input Stage

Webcam Capture

HD webcam captures live video stream in real-time at 30 FPS, providing continuous feed for face detection.

Frame Extraction

Individual frames are extracted from the video stream and preprocessed for optimal face detection accuracy.

Processing Stage

1. **Face Detection**
   Haar Cascade identifies face regions
2. **Feature Extraction**
   LBPH extracts facial features
3. **Face Recognition**
   Matches features with trained model
4. **Mask Detection**
   Checks for face mask presence

Output Stage

Display Results

Recognized person's name and confidence score displayed on screen with visual feedback.

Store Attendance

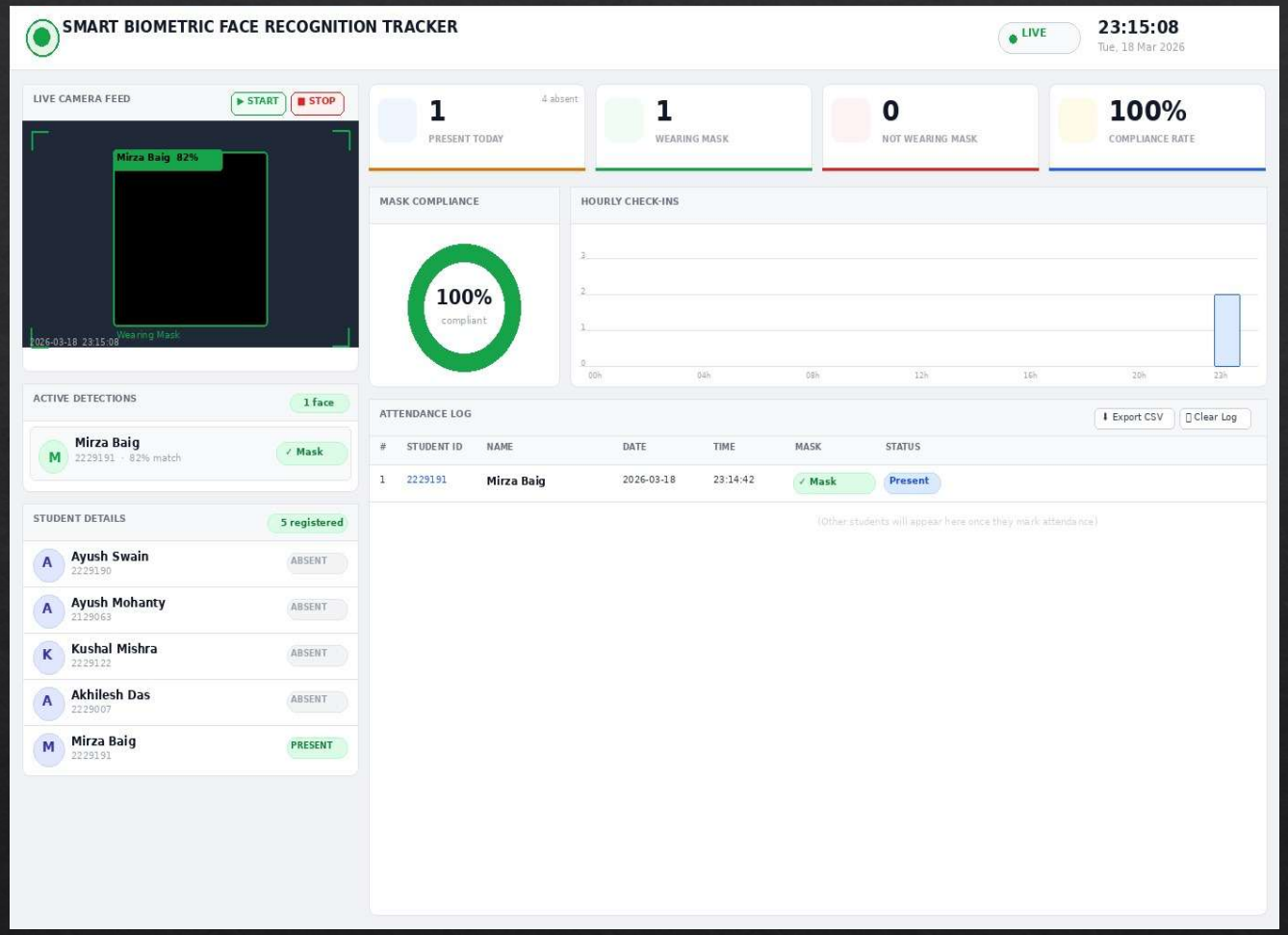
Attendance data saved to CSV file and MySQL database with timestamp for audit trail.

Pipeline Flow: Input → Processing → Output completes in <500ms for real-time performance

30 FPS	<500ms	Real-time
Frame Rate	Response Time	Processing

DASHBOARD INTERFACE

The Smart Biometric Face Recognition Tracker is an advanced system designed to automate attendance tracking while ensuring safety compliance through real-time facial recognition and mask detection. It captures live camera input, identifies individuals with high accuracy, and logs their attendance instantly along with details like time, date, and mask status. The dashboard provides a clear overview of daily presence, mask compliance rate, and hourly check-ins, making monitoring efficient and transparent. With features like active detections, student records, and exportable logs, the system reduces manual effort, minimizes errors, and enhances security—making it an ideal solution for institutions and workplaces aiming for smart, contactless, and reliable attendance management.



Dataset Creation

Image Capture Process

50 Images Per Person

Capture diverse angles, expressions, and lighting conditions to create a comprehensive training dataset that improves model generalization.

Organized Storage

Images stored in individual folders labeled with person ID for easy management and automated training pipeline integration.

Image Preprocessing Pipeline

01 Grayscale Conversion

Convert RGB images to grayscale to reduce computational complexity while preserving essential facial features for recognition.

02 Resize to 200×200

Standardize image dimensions to 200×200 pixels for consistent feature extraction and optimal model training performance.

03 Synthetic Mask Data

Generate masked versions of images to train the system for mask detection capability and ensure robustness in various scenarios.

04 Data Augmentation

Apply rotation, scaling, and brightness adjustments to increase dataset diversity and improve model robustness.

Dataset Statistics

Images Per Person	50
Image Resolution	200×200
Color Format	Grayscale
Masked Variants	Yes

Learning Outcome

This comprehensive dataset enables the system to learn facial features accurately, recognize individuals under various conditions, and detect mask presence with high reliability.

07 / Training

Model Training

Training Workflow

1. **Load Dataset Images**
   System reads all preprocessed images from dataset folders and organizes them with corresponding labels for supervised learning.
2. **Extract Features with LBPH**
   LBPH algorithm analyzes each image to extract unique facial features by comparing pixel intensity patterns in local neighborhoods.
3. **Train Model with Labels**
   Model learns to associate extracted features with person IDs, building a recognition database that maps features to identities.
4. **Save Trained Model**
   Trained model is saved as trainer.yml file, which contains all learned parameters and can be loaded for real-time recognition.

LBPH Algorithm

Local Binary Patterns Histograms is a robust face recognition algorithm that works by comparing each pixel with its neighbors and creating binary patterns that represent facial texture.

Robust Lighting invariant	**Fast** Real-time capable
Scalable Handles many faces	**Lightweight** Low resource usage

Learning Outcome

The trained model learns to recognize faces by creating a mathematical representation of each person's unique facial features, enabling accurate identification during real-time operation.

08 / Operation

Working Process

Real-Time Operation Flow

1. **Capture Live Video**
   Webcam continuously captures video feed at 30 FPS, providing real-time input for the face recognition pipeline.

2. **Detect Face**
   Haar Cascade classifier scans each frame to detect face regions and draws bounding boxes around detected faces.

3. **Recognize Face**
   Trained LBPH model extracts features from detected face and matches against the database to identify the person.

4. **Check Confidence**
   System evaluates confidence score to ensure reliable recognition. Only high-confidence matches are accepted.

Attendance Marking

If Match Found

- ✓ Display name on screen
- ✓ Mark attendance with timestamp
- ✓ Save to CSV file
- ✓ Store in MySQL database

Duplicate Prevention

System checks if attendance already marked for the day to prevent duplicate entries and maintain data integrity.

Performance Metrics

<500ms	95%+	30 FPS
Recognition Time	Accuracy	Frame Rate

09 / Conclusion

Conclusion & Future Scope

Project Achievements

Fast & Accurate System

Successfully developed a face recognition attendance system that processes recognition in under 500ms with 95%+ accuracy, significantly outperforming manual methods.

Contactless & Secure

Eliminated physical contact requirements, addressing post-pandemic health concerns while providing secure biometric verification that prevents proxy attendance.

Reduced Manual Errors

Automated the entire attendance process, eliminating human errors in data entry and saving 5-10 minutes of class time per session.

Future Enhancements

Multi-Face Recognition

Enable simultaneous recognition of multiple faces in a single frame for faster classroom attendance.

Mobile App Integration

Develop companion mobile application for remote monitoring and attendance management.

Cloud Storage

Implement cloud-based storage for centralized data management and backup capabilities.

Deep Learning Integration

Upgrade to CNN-based models for improved accuracy and advanced features like emotion detection.

Thank you for your attention

Questions & Discussion